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A co-evolutionary perspective on humans and *Mycobacterium tuberculosis* in the era of systems biology

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This Review Article explores the intricate relationship between humans and *Mycobacterium tuberculosis* (Mtb), providing an additional perspective on TB disease. Specifically, this review focuses on the utilization of systems-level approaches to study TB, while highlighting challenges in the frameworks used to identify the relevant immunologic signals that may explain the clinical spectrum of disease. The work could be further enhanced by better defining key terms that anchor the review, such as "unified mechanism" and "immunological route." This review will be of interest to immunologists as well as those interested in evolution and host-pathogen interactions.

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Abstract

Tuberculosis is once again the most fatal global infectious disease and has killed many more humans than any other pathogen. Despite the identification of *Mycobacterium tuberculosis* (Mtb) over 140 years ago, we have yet to control the epidemic. A central issue is the complexity of the host-pathogen interaction, with multiple underlying pathways leading to tuberculosis disease. This intricate relationship stems from the prolonged co-evolution of the pathogen with humans, resulting in diverse immunological processes leading to tuberculosis disease. Conversely, Mtb exposure may give a survival advantage through innate immune training, thereby providing selective pressure over millennia. Emerging methodologies, such as single cell and spatial transcriptomics, offer a golden opportunity to understand the immunology unpinning this host-pathogen interaction at unprecedented resolution. However, these analyses will be fundamentally flawed if they do not consider the intricacies of human Mtb infection. Here, we propose that attempts to find single immunological mechanisms leading to tuberculosis are hindering progress, and we must embrace the complexity of multiple paths to disease to allow the systems biology era to deliver transformative solutions.

Introduction

The study of tuberculosis (TB) pioneered infectious disease research in the modern scientific era, contributing to the formulation of Koch's postulates demonstrating that an illness can have an infectious origin (1). *Mycobacterium tuberculosis* (Mtb) infects and survives within macrophages, subverting the host immune response by multiple mechanisms including inhibition of phagosome

maturation and downregulation of antigen presenting molecules, leading to the formation of complex immune aggregates, known as granulomas (2). Even though TB was the first definitively identified infection, it remains the world's deadliest infectious disease despite well over 100 years of research (3). This contrast raises a critical question: why is *Mtb* proving so resistant to human efforts to control it?

Mtb has eluded attempts to develop a fully protective vaccine, despite a partially effective vaccine being available since the 1920's. *Mycobacterium bovis* BCG was developed by sequential culture of *M bovis* and protects children against disseminated TB but has limited protection against adult disease (4, 5). Although exciting progress has been made with a vaccine that reduces progression to overt TB disease by 50% when given to those with immunological evidence of latent infection in a phase II study (6), currently the immune mechanisms underpinning protection have not been identified. Subsequently, a major trial of an alternative vaccine showed no efficacy in preventing recurrence. Indeed, despite being highly immunogenic, the relapse rate tended to be higher in the vaccine group (7). Similarly, repeat BCG vaccination does not increase protection, despite inducing a strong *Mtb*-specific CD4 T cell response (8).

These difficulties highlight the priority of understanding the host-pathogen interaction more fully. We have insufficient knowledge of key steps in disease progression to develop transformative interventions. Heterogeneity across the spectrum of human TB is well described (9, 10), but the majority of fundamental investigations into disease mechanisms are based on the premise of a consistent underlying process, and this can be understood through reductionist scientific approaches. However, clinical observations demonstrate that there are multiple paths to TB, and so seeking to define a single mechanism is likely to be flawed.

Immunological insights from historic and recent clinical observations

TB has co-evolved with humans for millennia, with some estimates suggesting up to 70,000 years (11), though other analyses suggest the most recent common ancestor was ~6,000 years before present (12, 13). The field of paleoarchaeology provides extensive evidence of TB from the early neolithic period in the Middle East, with approximately 5% of skeletons from a 10,000-year-old village showing signs of TB (14). This was just before animal domestication and pottery, in huntergatherers who built stone houses, and so *Mtb* was already successfully transmitting in humans before the subsequent population growth that occurred with farming (14). Potentially, to survive in relatively small hunter-gatherer communities *Mtb* may have needed to have reduced virulence to avoid excessive deaths and a latent period to permit sustainable transmission in low population numbers (15). With the expansion and increased density of human populations, more rapid progression to TB disease can be sustained, consistent with analysis that most TB progression in high incidence settings occurs within 1–2 years of exposure (16).

Mtb successfully persisted over the ages and then flourished in the crowded populations that occurred with the industrial revolution (17), giving rise to the modern TB era approximately 250 years ago. The fundamental cause of TB remained unknown until Koch's seminal work (1). Early investigations demonstrated that the first *Mtb* infection point was the lung base, while *Mtb* exits from the apices of the lungs (18). This life cycle must involve several distinct host/pathogen interactions, as initially immune evasion is required for *Mtb* to survive, but then later immune engagement is necessary to cause the inflammation and lung destruction needed to optimise transmission (19). Cavitory lung disease leads to proliferation of extracellular bacteria and increased transmission (20). Notably, most people (approximately 90%) initially infected with *Mtb* never progress to active, clinical disease (3). In addition, in the pre-antibiotic era, the progression and regression of different lesions in the same individual were observed on chest radiographs, and one third of patients with active TB disease self-healed, showing that the host/pathogen interaction is finely balanced at all stages of infection (17).

Modern immunological techniques and the development of biologic therapies that target specific immune processes have provided extensive insight into TB disease mechanisms. The greatly increased occurrence of TB in the context of HIV co-infection, for example, highlighted immunodeficiency as a major driver of disease (21). Similarly, the occurrence of TB after anti-TNF- α therapy for autoimmune disease confirmed the importance of TNF- α in control of latent infection (22). Furthermore, genetic investigations have identified numerous immunodeficiencies via studies of Mendelian Susceptibility to Mycobacterial Diseases (MSMD), with mutations typically along the IL-12/IFN- γ /STAT signalling pathway (23–25). With less clearly defined immunologic mechanisms, malnutrition is a significant risk factor for TB (26), and food supplementation reduces TB incidence in contacts (27). Therefore, diverse immune deficiencies can lead to active TB.

The vast majority of patients who develop TB, however, have no clear identifiable immunodeficiency. Indeed, Comstock's seminal study from the 1970s showed that children from Haiti with the strongest recall responses to Mtb antigens actually had the greatest subsequent risk of developing TB (28). These observations have been replicated in more recent studies using IFN- γ release assays (IGRAs) in response to TB antigens, where higher IFN- γ production associates with increased risk of progressing to disease in both children and adults (29, 30). TB is commonest in young adults in their immunological prime, and more frequent in males than females, characterised by an excessive inflammatory response (31). The implication that TB can also be caused by immune excess is now supported by recently introduced cancer immunotherapies (32). Anti-PD-1 treatment, which activates the immune response and represents the immunological opposite to anti-TNF therapy, should control TB if immunodeficiency were the critical component. However, anti-PD-1 treatment can lead to rapid reactivation of latent TB infection, first identified in case reports (33). This finding is supported by studies in mice (34, 35), the non-human primate (36) and 3D cellular models (37), and ultimately has been validated by patient registry studies (38, 39). Similarly, type II diabetes is associated with an increased risk of TB (26), characterized by a hyper-inflammatory immune response (40). Therefore, diverse clinical evidence demonstrates that there are multiple immunological disturbances that can lead to TB disease (41) (Figure 1 [↗](#)).

Even when TB develops in the face of a “normal” immune system, there are likely to be many different subgroups that have not yet been identified due to limitations in standard immunological profiling. A quarter of the world's population are thought to be exposed to Mtb (42), and so the vast majority of those infected with Mtb remain healthy lifelong (3), while those who progress to active TB likely represent distinct outlier populations. The diverse causes of active TB, such as anti-TNF, MSMD, HIV, diabetes and anti-PD1, demonstrate that patients do not reach TB by a single pathway, but instead lose the immunological balance that controls Mtb by multiple paths. These observations may explain why genetic studies have generally failed to find consistent predispositions. Evidence of heritability can be demonstrated but is often hard to validate in different populations (43). Mutations in the macrophage endosomal protein NRAMP1, for example, were shown to associate with TB in an early seminal study (44), but since then few consistent traits of genetic susceptibility to TB have been identified (45).

Recent genomic studies demonstrate the complex co-evolution of host and Mtb. The International Tuberculosis Host Genetics Consortium's (ITHGC) first analysis found one significant host genetic variant, human leukocyte antigen (HLA)-II region (rs28383206), which conferred TB susceptibility across nine genome-wide association studies (GWAS) across three continents (43). However, other variants previously associated with TB susceptibility were not replicated. Another approach using genome-to-genome analysis of paired human and Mtb genomes in Peru identified another determinant on chromosome 6, rs3130660, in the flotillin-1 (FLOT1) gene (46). Together with our understanding of the co-adaptation of different Mtb lineages with human migrations (47), the host immune response to Mtb is likely to be dependent on ancestral-related genetic factors and complex host-pathogen dynamics which remain incompletely understood.

Studies of individuals who resist Mtb infection despite recurrent exposure generates a diverse list of potentially protective features, including T cell subsets or activation (48–50), TNF- α responses (51), antibody activity (52) and innate immune training (53). These human observations are

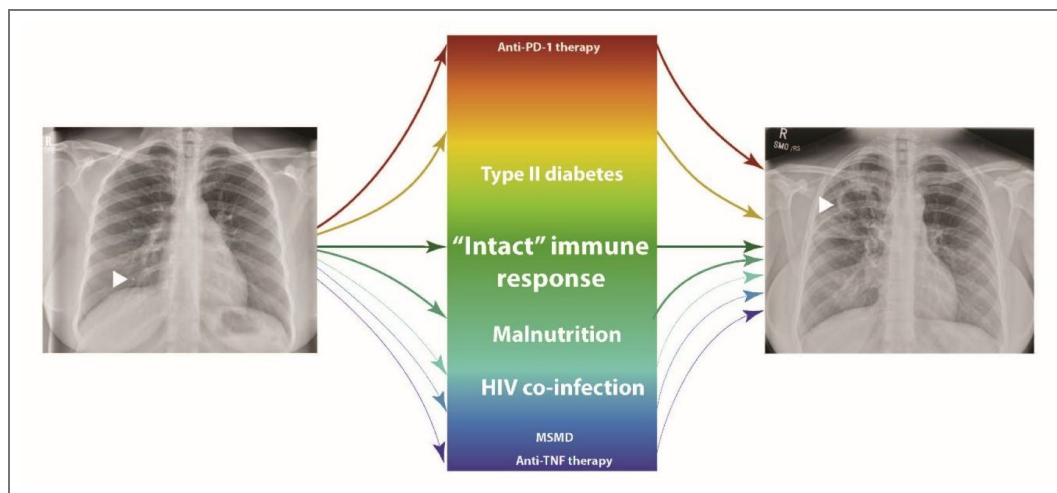


Figure 1. Multiple pathways can lead to TB disease, from opposing immunological extremes.

Generally, these can be viewed as immune deficiencies or immune excess, but the majority of patients who develop TB are relatively young with a competent immune response, illustrating the complexity of this spectrum. Though not quantitative, font size relates to contribution to global incidence. Left arrow: Ghon focus at lung base; Right arrow: cavity at lung apex.

supported by murine genetic experiments, which show that increased susceptibility to Mtb can result from a wide range of immunological alterations (54). Similarly, infant vaccine studies show that there are distinct patterns of response, that may determine vaccine efficacy (55). However, despite this evidence of diversity, the majority of fundamental studies continue to seek a single underlying mechanism that leads to TB disease progression, which is incompatible with clinical observations.

Ultimately, to transmit efficiently, Mtb needs to cause pulmonary disease to then spread by airborne droplets (20), and for Mtb it does not matter the route taken, as long as it ends at pulmonary TB. Increasing evidence suggests that asymptomatic transmission may be important in high incidence settings, potentially in the absence of overt pulmonary disease (56–58). Experimentally, TB is a fundamentally challenging disease to model, as the interaction is prolonged and Mtb is an obligate human pathogen (59). The primary driver of advances in immunology in the last decades has been transgenic mice (60), but the mouse model of TB does not accurately reflect human disease (61). Disease heterogeneity has been highlighted in describing clinical TB endotypes observed during active disease (62), with different endotypes exhibiting diverse immunological characteristics and association with outcome (63). However, we propose that insufficient attention has been given to the different immunological pathways that may converge on the same disease phenotype.

Mtb's single successful establishment in humanity

Just as new tools are providing insights into the complexity of human TB progression, advances in mycobacterial genomics are highlighting the unique nature of the human-Mtb relationship (64). Mtb is a near clonal organism, with evidence suggesting that there has been only one successful and sustained penetration into the human population (47, 64, 65). The entire spectrum of Mtb strains globally only differs by a total of approximately 2,000 SNPs (64). This genetic conservation has persisted from the most recent common ancestor (66) during the expansion of Mtb in human populations since the industrial revolution, which created much denser human aggregations suitable for transmission (17). This suggests that Mtb was already close to being optimised for human hosts. Further evolution may have been to increase transmission within specific populations as humans diverged genetically (65), with a recent study suggesting that in fact Mtb may be becoming attenuated to increase spread in populations (67).

A very similar organism, *M. canettii*, can cause disease but cannot transmit from human to human (68). Similarly, *M. bovis* is 99.9% identical to Mtb (69), but has never achieved sustained human-to-human transmission, despite many millions of human exposures during the prepasteurisation era and common lymph node infections (65). Therefore, human TB is caused exclusively by Mtb, unlike other very closely related mycobacteria, which maintain infection cycles in other higher mammals but not humans (65). Clues to the key pathogenic mechanisms may lie in the differences between mycobacterial species (70), but ultimately similarities between Mtb strains must be critical to their ongoing success. For example, the hyper conservation of T cell epitopes may be evidence that Mtb manipulates the host immune response to favour disease and transmission (71). Given the size of the Mtb genome, identifying the critical conserved features will be challenging, especially as half of its genes still lack a known function 25 years after it was first sequenced (72).

One highly intriguing proposition is that latent TB may itself give humans an evolutionary advantage (72). Exposure to Mtb modifies innate immune training via reprogramming hematopoietic stem cells (73), and so a mechanism whereby Mtb could protect from other fatal infections is plausible. Similarly, *M. bovis* BCG, a live-attenuated strain of *M. bovis*, causes innate immune training (74), suggesting this effect is common to Mtb and BCG. BCG vaccination reduces mortality much more significantly than can be explained by the effect on TB incidence alone (75), with a protective effect confirmed in diverse studies (76). For example, BCG reduces viral infections in infants in Africa (77) and experimentally challenged adults (78), and associates with improved survival in Europe (79). Mtb and BCG can protect against SARS-CoV2 infection (80, 81), although this did not translate to efficacy in a clinical trial (82). Together, these observations strongly imply that mycobacterial infection protects from other infectious causes of death.

Mtb almost certainly first became established in humans in East Africa (47, 65), potentially about 70,000 years ago (11), although this timeline is debated. Several human migrations out of Africa are thought to have preceded this date, but all ultimately became extinct, with the first sustained human dispersal 60 – 70,000 years ago (83, 84). *Mtb* diversity mirrors human mitochondrial genome diversity, further implying that *Mtb* disseminated with human populations from East Africa (47). The primary selective pressure in these early communities would have been infectious disease (85). This raises a novel hypothesis that a survival advantage for the first successful human migrants out of Africa was *Mtb* circulating in the community, reducing mortality from other infectious diseases and thereby enabling sustainable population growth.

Mtb transmission may have benefitted humans by increasing innate immune resistance to infection at the cost of 10% disease penetrance that permits *Mtb* propagation. This selective pressure over many millennia would progressively remove genotypes that lead to high susceptibility to TB, but equally would select against individuals with complete resistance to initial *Mtb* infection (Figure 2). This could explain why consistent genetic traits for susceptibility or resistance to TB have been hard to identify (45). More recent mass infection events, such as the smallpox epidemics that killed approximately 25% of the population (85), may have further favoured individuals immunologically trained by *Mtb*. Likewise, successive waves of plague killed approximately 25% of the population, at the end of the Roman empire and then in the European middle ages, adding to selective pressure from endemic infections (86, 87). If humans have been selected to be permissive to *Mtb* infection but resistant to TB disease, which could be regarded as colonisation, it suggests the development of active disease must be a relatively unusual event in a subset of outlier individuals. In some sense, we could be regarded as having a symbiotic relationship with *Mtb*, with disease representing a necessary evil, caused by an imbalance in the predominantly stable host pathogen interaction (88, 89).

The host-pathogen interaction at a cellular level

The early histological era described the wide range of human TB lesions and granuloma types, and identified TB as a disease characterised by spatial organisation (90). Classically, the granuloma has been proposed to be where the outcome of the host-pathogen interaction is determined (91). Recent methodological advances are permitting much greater dissection of events and further highlight the importance of spatial organisation within the granuloma (92–94). However, just as the early X-ray era showed some lesions progressing and some regressing, these studies demonstrate the great heterogeneity between granuloma types. Studies in the nonhuman primate have allowed investigation into features of progressing and controlling granulomas, identifying potential correlates of immune control (95), but even this model only partially recapitulates human disease.

Furthermore, the recent spatial studies highlight the complexity of cellular players, including the established fulcrum of macrophages and T cells, but additionally the importance of B cells, neutrophils and fibroblasts. For example, fibroblasts are emerging as important immune regulators in other lung diseases (96), and so it seems highly likely that they play an active role in TB-related inflammation. Fibroblast zonation can lead to feed-forward inflammatory loops and so may propagate disease (97). Consequently, the full spectrum of cell types in *Mtb*-infected lesions needs to be considered. Given that multiple underlying immunological pathways can lead to active TB, it seems unlikely that a single cellular component will fully explain the balance between *Mtb* containment and progression to active disease. However, the majority of studies continue to look for a single consistent immune mechanism; discussions rarely state “this is one of several potential routes to TB disease” (49, 95, 98–100).

Considering the clinical and experimental evidence, immunological failure is likely to be a multistep process, whereby either one large deficit, or numerous small imbalances, can lead to progression and disease (Figure 3). *Mtb* and humans interact over many years, as the pathogen is difficult to eradicate due to its highly evolved survival mechanisms (101), and therefore in those individuals in whom it survives, there is a long period where it can escape host control. Potentially these different paths may ultimately cross or converge; if so, understanding the key nodes will

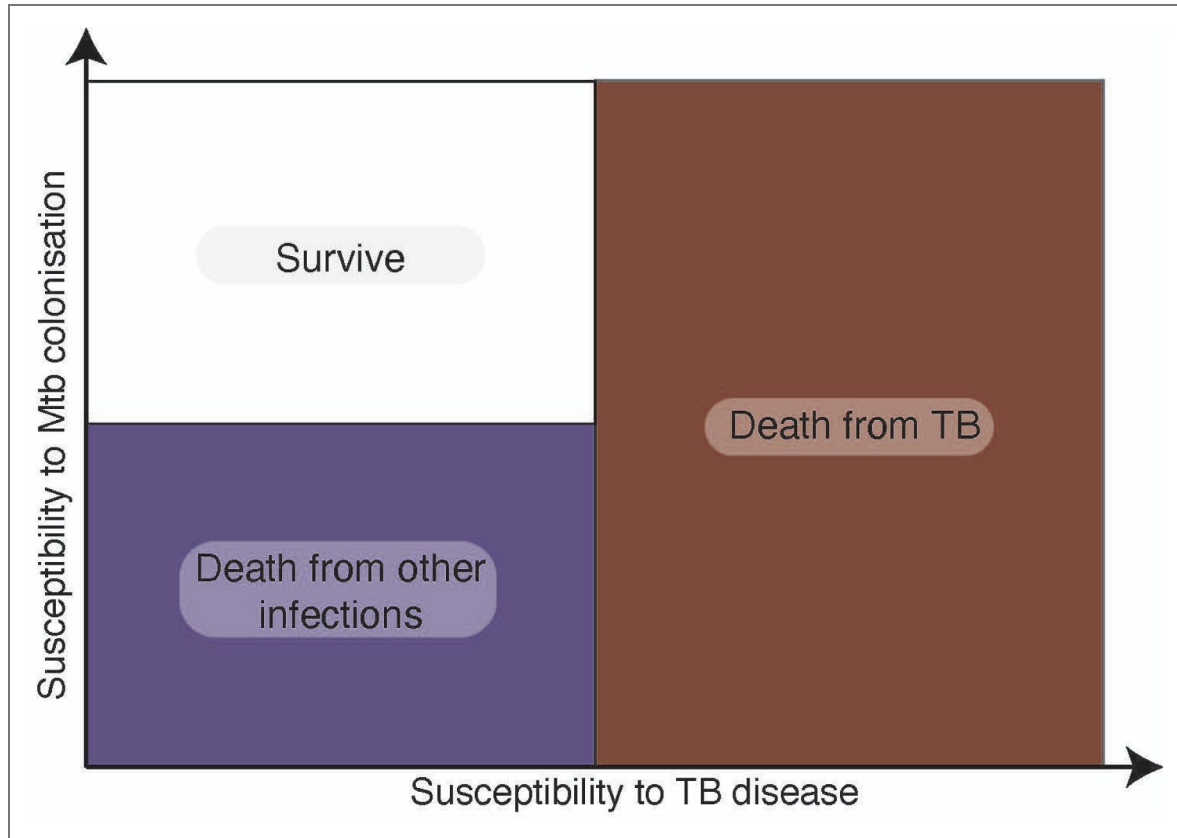


Figure 2. Selection pressure of prolonged co-evolution favours individuals permissive to asymptomatic Mtb colonisation but resistant to active disease.

Over millennia, Mtb circulation in society will remove genetic traits that cause high susceptibility to active TB infection. Perhaps less intuitively, if Mtb generates trained immunity that protects against other fatal diseases, individuals with low susceptibility to initial Mtb infection will also be selected against due to increased mortality from other infections. The resulting population would then reflect modern humans; highly susceptible to initial Mtb colonisation but with low susceptibility to TB disease.

allow more broadly effective treatments to emerge. For example, lung extracellular matrix degradation could be regarded as a final common pathway (102), but, again, this may result from different collagenases including macrophage-derived MMP-1 or neutrophil-derived MMP-8 (103, 104).

Emerging methodologies and the challenge of data analysis

The recent adoption of “omic” methodologies, including single cell transcriptomics, spatial transcriptomics, and proteomics, offers unprecedented opportunities to dissect the mechanisms of human TB pathogenesis and accelerate the development of effective interventions. These approaches generate large-scale, complex datasets that capture the heterogeneity of TB lesions. However, this complexity also presents significant analytical challenges. Standard computational pipelines, if not adapted to account for the biological and technical variability in these data, are unlikely to deliver robust or reproducible insights. A major obstacle is the integration of data from multiple studies and platforms, which can differ both within a single omic layer (horizontal integration) and across multiple omic layers (vertical integration) (105). Such integration risks data loss and inconsistent results, especially when data are not harmonised.

The prolonged co-evolution between host and pathogen has resulted in multiple immunological pathways to active TB, significantly adding to the biological heterogeneity that complicates the analysis and interpretation of multi-omic data. Addressing this complexity requires well-annotated clinical cohorts that capture the full spectrum of TB heterogeneity, ideally with longitudinal outcome data rather than single time-point snapshots. Comprehensive clinical descriptors and associated data such as laboratory analyses and chest X-rays will permit definition of each clinical phenotype. To accommodate the diversity of disease pathways, novel bioinformatic approaches are needed that move beyond the assumption of a single sequence of events that results in active TB.

The power of multi-omic approaches to reveal complex molecular relationships depends on both the quality of the omic data and the fit between experimental design and data integration strategies. For example, correlation-based integration requires matched samples across omics, sufficient sample numbers, and comparable variance structures. These requirements are often overlooked, leading to insufficient power, noisy data, and unrealistic integration (106). Each omic technology brings its own challenges. Signal-to-noise ratios differ across modalities, and appropriate algorithms are needed to estimate sample size and power for each. Notably, using the same sample numbers across modalities does not ensure comparable statistical power; achieving equal power can require unbalanced sample sizes (107). Other recognised challenges include the interpretation and validation of multi-omic models, standardised annotation, and data sharing (107). Recent developments include web-based, user-friendly tools that enable both knowledge- and data-driven multi-omic integration (108). Importantly, a wider use of artificial intelligence (AI) is transforming omics data analysis by providing robust methods to interpret complex biological datasets (109). Machine learning and deep learning techniques are now routinely applied to DNA, transcriptomic, proteomic, and metabolomic data, enabling more integrated and comprehensive analyses (109). In proteomics, for example, AI-driven approaches have enhanced peptide measurement predictions and accelerated biomarker discovery, often outperforming conventional assays (110). To improve interpretability, explainable AI (XAI) methods are increasingly employed, with feature relevance mapping and visual explanations emerging as preferred post hoc strategies (111). However, despite these developments, significant challenges remain in implementing XAI. Further research is needed to overcome these barriers and unlock the full translational potential of AI in omics analysis (110, 111). Ultimately, the utility of these innovations will depend on the functional validation of the resulting models.

Given the complexity of human TB, careful study design and unbiased integration methods that accommodate data limitations are essential. Spatial context is particularly important, as TB pathology involves three-dimensional immune responses. Extracellular matrix remodelling is a

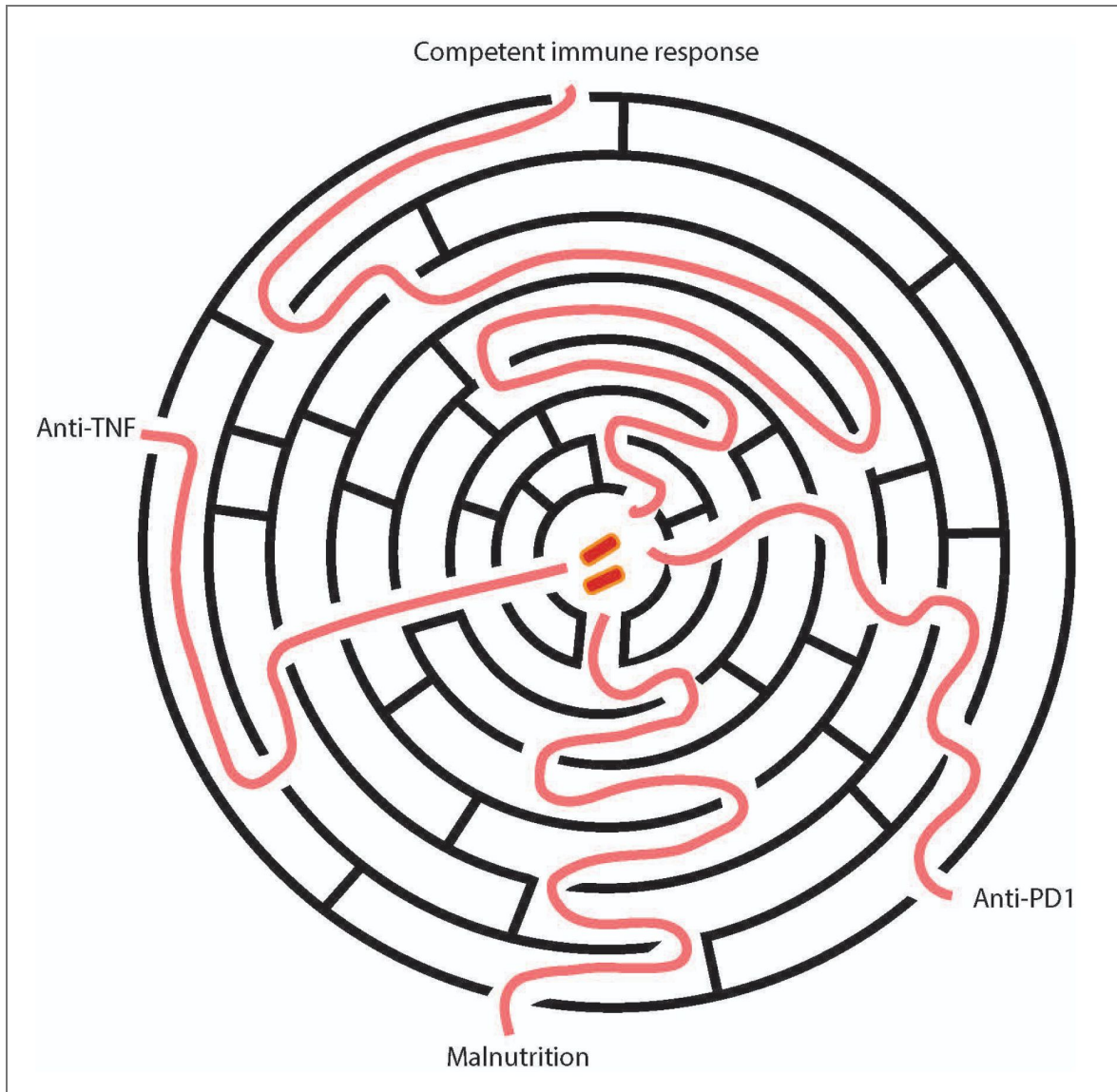


Figure 3. Schematic of the interactions needed for Mtb to escape the host immune response.

As control of Mtb requires a co-ordinated host response, there are multiple sequences of immune events that can ultimately result in progression to active TB disease. A major immune disturbance, such as TNF- α or PD-1 inhibition, gives a relatively direct pathway to active TB. However, most individuals develop TB due to a series of less apparent immune events and no clear global immune disturbance that can be identified by current immune profiling approaches.

hallmark of TB granulomas [81], and the matrix regulates host cell biology [84]. Consequently, multiple data inputs such as matrix composition and organisation may be needed alongside host and Mtb transcriptomic data. Ultimately, local cellular events must be modelled at the tissue level, providing a second layer of computational complexity [85].

Addressing these challenges will require substantial investment in analytical capacity. Multi-omics is already transforming our understanding of disease heterogeneity, facilitating the identification of previously unrecognized subgroups, refining prognostic and therapeutic approaches, and providing deeper mechanistic insights. This strategy has successfully stratified rare tumours (112), profiled healthy populations (113), and enabled health screening to reveal previously hidden disease and risk subgroups (114), supporting the shift toward precision medicine. As human disease processes are rarely uniform, advances in multi-omic study design and analysis for TB will likely benefit a broad range of conditions. Ultimately, this comprehensive understanding of disease pathophysiology can then lead to more targeted and individualised treatment, although implementation will require developments in companion diagnostics to accurately stratify patients.

Implications of diverse disease pathways for new TB interventions

Ultimately, the complexity of human TB underlies our inability to deploy a transformative intervention, and so global mortality remains depressingly high. The human-Mtb interaction is so closely co-evolved that experimental findings need to be interpreted in light of human disease phenomena. Multiomic studies in TB are currently being undertaken on small numbers due to cost and challenges in obtaining appropriate clinical samples, in specific regions, and so are unlikely to reflect the global heterogeneity. Therefore, results need to be interpreted with caution and wider studies will be needed to confirm the generalisability of findings. A critical aspect will be carefully curated and fully accessible metadata, so that as the body of omic datasets on human TB increase, they can be accurately integrated into wider analyses. Recurrent mining of these datasets is likely to be fundamental to understanding the breadth of TB pathogenesis. Missing metadata makes interpretation difficult, and in worst cases misleading. In addition, innovative computational approaches will be required whereby the analysis specifically accommodates multiple immune pathways to disease.

Given the toll that TB takes in the poorest parts of the world, we have a moral imperative to end the epidemic (115). To achieve this, we must acknowledge the complexity of human TB that has resulted from our prolonged co-evolution with the pathogen and the selective pressure of persistent Mtb exposure over millennia. Success depends on integrating the full spectrum of TB disease into our bioinformatic analyses, and ultimately understanding TB's complexity can then inform logical interventions. If we seek a single mechanistic explanation of TB disease, this seems to be unlikely to be successful.

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Peer reviews

Reviewer #1 (Public review):

Summary:

This is an interesting and useful review highlighting the complex pathways through which pulmonary colonisation or infection with *Mycobacterium tuberculosis* (Mtb) may progress to develop symptomatic disease and transmit the pathogen. I found the section on immune correlates associated with individuals who have clearly been exposed to and reacted to Mtb

but did not develop latent infections particularly valuable. However, several aspects would benefit from clarification.

Strengths:

The main strengths lie in the arguments presented for a multiplicity of immune pathways to TB disease.

Weaknesses:

The main weaknesses lie in clarity, particularly in the precise meanings of the three figures.

I accept that there is a 'goldilocks zone' that underpins the majority of TB cases we see and predominantly reflects different patterns of immune response, but the analogies used need to be more clearly thought through.

<https://doi.org/10.7554/eLife.108175.2.sa1>

Reviewer #2 (Public review):

Summary:

This is a thought-provoking perspective by Reichmann et al, outlining supportive evidence that *Mycobacterium tuberculosis* co-evolved with its host *Homo Sapiens* to both increase susceptibility to infection and reduce rates of fatal disease through decreased virulence. TB is an ancient disease where two modes of virulence are likely to have evolved through different stages of human evolution: one before the Neolithic Demographic Transition, where humans lived in sparse hunter-gatherer communities, which likely selected for prolonged Mtb infection with reduced virulence to allow for transmission across sparse populations. Conversely, following the agricultural and industrial revolutions, Mtb virulence is likely to have evolved to attack a higher number of susceptible individuals. These different disease modalities highlight the central idea that there are different immunological routes to TB disease, which converge on a disease phenotype characterized by high bacterial load and destruction of the extracellular matrix. The writing is very clear and provides a lot of supportive evidence from population studies and the recent clinical trials of novel TB vaccines, like M72 and H56. However, there are areas to support the thesis that have been described only in broad strokes, including the impact of host and Mtb genetic heterogeneity on this selection, and the alternative model that there are likely different TB diseases (as opposed to different routes to the same disease), as described by several groups advancing the concept of heterogeneous TB endotypes. I expand on specific points below.

Strengths:

(1) The idea that Mtb evolved to both increase transmission (and possible commensalism with humans) with low rates of reactivation is intriguing. The heterogeneous TB phenotypes in the collaborative cross model (PMID: 35112666) support this idea, where some genetic backgrounds can tolerate a high bacterial load with minimal pathology, while others show signs of pathogenesis with low bacterial loads. This supports the idea that the underlying host state, driven by a number of factors like genetics and nutrition, is likely to explain whether someone will co-exist with Mtb without pathology, or progress to disease. I particularly enjoyed the discussion of the protective advantages provided by Mtb infection, which may have rewired the human immune system to provide protection against heterologous pathogens- this is supported by recent studies showing that Mtb infection provides moderate protection against SARS-CoV-2 (PMID: 35325013, and 37720210), and may have applied to other viruses that are likely to have played a more significant role in the past in the natural selection of *Homo Sapiens*.

(2) Modeling from Marcel Behr and colleagues (PMID: 31649096) indeed suggests that there are at least TB clinical phenotypes that likely mirror the two distinct phases of Mtb co-evolution with humans. Most of the TB disease progression occurs rapidly (within 1-2 years of exposure), and the rest are slow cases of reactivation over time. I enjoyed the discussion of the difference between the types of immune hits needed to progress to disease in the two scenarios, where you may need severe immune hits for rapid progression, a phenotype that likely evolved after the Neolithic transition to larger human populations. On the other hand, a series of milder immune events leading to reactivation after a long period of asymptomatic infection likely mirrors slow progression in the hunter-gatherer communities, to allow for prolonged transmission in scarce populations. Perhaps a clearer analysis of these models would be helpful for the reader.

Weaknesses:

(1) The discussion of genetic heterogeneity is limited and only discusses evidence from MSMD studies. Genetics is an important angle to consider in the co-evolution of Mtb and humans. There is a large body of literature on both host and Mtb genetic associations with TB disease. The very fact that host variants in one population do not necessarily cross-validate across populations is evidence in support of population-specific adaptations. Specific Mtb lineages are likely to have co-evolved with distinct human populations. A key reference is missing (PMID: 23995134), which shows that different lineages co-evolved with human migrations. Also, meta-analyses of human GWAS studies to define variants associated with TB are very relevant to the topic of co-evolution (e.g., PMID: 38224499). eQTL studies can also highlight genetic variants associated with regulating key immune genes involved in the response to TB. The authors do mention that Mtb itself is relatively clonal with ~2K SNPs marking Mtb variation, much of which has likely evolved under the selection pressure of modern antibiotics. However, some of this limited universe of variants can still explain co-adaptations between distinct Mtb lineages and different human populations, as shown recently in the co-evolution of lineage 2 with a variant common in Peruvians (PMID: 39613754).

(2) Although the examples of anti-TNF and anti-PD1 treatments are relevant as drivers of TB in limited clinical contexts, the bigger picture is that they highlight major distinct disease endotypes. These restricted examples show that TB can be driven by immune deficiency (as in the case of anti-TNF, HIV, and malnutrition) or hyperactivation (as in the case of anti-PD1 treatment), but there are still certainly many other routes leading to immune suppression or hyperactivation. Considering the idea of hyper-activation as a TB driver, the apparent higher rate of recurrence in the H56 trial referenced in the review is likely due to immune hyperactivation, especially in the context of residual bacteria in the lung. These different TB manifestations (immune suppression vs immune hyperactivation) mirror TB endotypes described by DiNardo et al (PMID: 35169026) from analysis of extensive transcriptomic data, which indicate that it's not merely different routes leading to the same final endpoint of clinical disease, but rather multiple different disease endpoints. A similar scenario is shown in the transcriptomic signatures underlying disease progression in BCG-vaccinated infants, where two distinct clusters mirrored the hyperactivation and immune suppression phenotypes (PMID: 27183822). A discussion of how to think about translating the extensive information from system biology into treatment stratification approaches, or adjunct host-directed therapies, would be helpful.

<https://doi.org/10.7554/eLife.108175.2.sa2>

Reviewer #3 (Public review):

Summary:

This perspective article by Reichmann et al. highlights the importance of moving beyond the search for a single, unified immune mechanism to explain host-Mtb interactions. Drawing from studies in immune profiling, host and bacterial genetics, the authors emphasize inconsistencies in the literature and argue for broader, more integrative models. Overall, the article is thought-provoking and well-articulated, raising a concept that is worth further exploration in the TB field.

Strengths:

Timely and relevant in the context of the rapidly expanding multi-omics datasets that provide unprecedented insights into host-Mtb interactions.

Weaknesses (Minor):

(1) Clarity on the notion of a "unified mechanism". It remains unclear whether prior studies explicitly proposed a single unifying immunological model. While inconsistencies in findings exist, they do not necessarily demonstrate that earlier work was uniformly "single-minded". Moreover, heterogeneity in TB has been recognized previously (PMIDs: 19855401, 28736436), which the authors could acknowledge.

(2) Evolutionary timeline and industrial-era framing. The evolutionary model is outdated. Ancient DNA studies place the Mtb's most recent common ancestor at ~6,000 years BP (PMIDs: 25141181; 25848958). The Industrial Revolution is cited as a driver of TB expansion, but this remains speculative without bacterial-genomics evidence and should be framed as a hypothesis. Additionally, the claim that Mtb genomes have been conserved only since the Industrial Revolution (lines 165-167) is inaccurate; conservation extends back to the MRCA (PMID: 31448322).

(3) Trained immunity and TB infection. The treatment of trained immunity is incomplete. While BCG vaccination is known to induce trained immunity (ref 59), revaccination does not provide sustained protection (ref 8), and importantly, Mtb infection itself can also impart trained immunity (PMID: 33125891). Including these nuances would strengthen the discussion.

<https://doi.org/10.7554/eLife.108175.2.sa3>

Author response:

The following is the authors' response to the original reviews.

eLife Assessment

This Review Article explores the intricate relationship between humans and Mycobacterium tuberculosis (Mtb), providing an additional perspective on TB disease. Specifically, this review focuses on the utilization of systems-level approaches to study TB, while highlighting challenges in the frameworks used to identify the relevant immunologic signals that may explain the clinical spectrum of disease. The work could be further enhanced by better defining key terms that anchor the review, such as "unified mechanism" and "immunological route." This review will be of interest to immunologists as well as those interested in evolution and host-pathogen interactions.

We thank the editors for reviewing our article and for the primarily positive comments. We accept that better definition and terminology will improve the clarity of the message, and so have changed the wording as suggested above in the revised manuscript.

Public Reviews:

Reviewer #1 (Public review):
Summary:

This is an interesting and useful review highlighting the complex pathways through which pulmonary colonisation or infection with Mycobacterium tuberculosis (Mtb) may progress to develop symptomatic disease and transmit the pathogen. I found the section on immune correlates associated with individuals who have clearly been exposed to and reacted to Mtb but did not develop latent infections particularly valuable. However, several aspects would benefit from clarification.

Strengths:

The main strengths lie in the arguments presented for a multiplicity of immune pathways to TB disease.

Weaknesses:

The main weaknesses lie in clarity, particularly in the precise meanings of the three figures.

We accept this point, and have completely changed figure 2, and have expanded the legends for figure 1 and 3 to maximise clarity.

I accept that there is a 'goldilocks zone' that underpins the majority of TB cases we see and predominantly reflects different patterns of immune response, but the analogies used need to be more clearly thought through.

We are glad the reviewer agrees with the fundamental argument of different patterns of immunity, and have revised the manuscript throughout where we feel the analogies could be clarified.

Reviewer #2 (Public review):
Summary:

This is a thought-provoking perspective by Reichmann et al, outlining supportive evidence that Mycobacterium tuberculosis co-evolved with its host Homo Sapiens to both increase susceptibility to infection and reduce rates of fatal disease through decreased virulence. TB is an ancient disease where two modes of virulence are likely to have evolved through different stages of human evolution: one before the Neolithic Demographic Transition, where humans lived in sparse hunter-gatherer communities, which likely selected for prolonged Mtb infection with reduced virulence to allow for transmission across sparse populations. Conversely, following the agricultural and industrial revolutions, Mtb virulence is likely to have evolved to attack a higher number of susceptible individuals. These different disease modalities highlight the central idea that there are different immunological routes to TB disease, which converge on a disease phenotype characterized by high bacterial load and destruction of the extracellular matrix. The writing is very clear and provides a lot of supportive evidence from population studies and the recent clinical trials of novel TB vaccines, like M72 and H56. However, there are areas to support the thesis that have been described only in broad strokes, including the impact of host and Mtb genetic heterogeneity on this selection, and the alternative model that there are likely different TB diseases (as opposed to different routes to the same disease), as described by several groups advancing the concept of heterogeneous TB endotypes. I expand on specific points below.

Strengths:

The idea that Mtb evolved to both increase transmission (and possible commensalism with humans) with low rates of reactivation is intriguing. The heterogeneous TB phenotypes in the collaborative cross model (PMID: 35112666) support this idea, where some genetic backgrounds can tolerate a high bacterial load with minimal pathology, while others show signs of pathogenesis with low bacterial loads. This supports the idea that the underlying host state, driven by a number of factors like genetics and nutrition, is likely to explain whether someone will co-exist with Mtb without pathology, or progress to disease. I particularly enjoyed the discussion of the protective advantages provided by Mtb infection, which may have rewired the human immune system to provide protection against heterologous pathogens- this is supported by recent studies showing that Mtb infection provides moderate protection against SARS-CoV-2 (PMID: 35325013, and 37720210), and may have applied to other viruses that are likely to have played a more significant role in the past in the natural selection of Homo Sapiens.

We thank the reviewer for their positive comments, and also for pointing out work that we have overlooked citing previously. We now discuss and cite the work above as suggested

Modeling from Marcel Behr and colleagues (PMID: 31649096) indeed suggests that there are at least TB clinical phenotypes that likely mirror the two distinct phases of Mtb co-evolution with humans. Most of the TB disease progression occurs rapidly (within 1-2 years of exposure), and the rest are slow cases of reactivation over time. I enjoyed the discussion of the difference between the types of immune hits needed to progress to disease in the two scenarios, where you may need severe immune hits for rapid progression, a phenotype that likely evolved after the Neolithic transition to larger human populations. On the other hand, a series of milder immune events leading to reactivation after a long period of asymptomatic infection likely mirrors slow progression in the hunter-gatherer communities, to allow for prolonged transmission in scarce populations. Perhaps a clearer analysis of these models would be helpful for the reader.

We agree that we did not present these concepts in as much detail as we should, and so we now discuss this more on lines 81 – 83 and 184 - 187)

Weaknesses:

The discussion of genetic heterogeneity is limited and only discusses evidence from MSMD studies. Genetics is an important angle to consider in the co-evolution of Mtb and humans. There is a large body of literature on both host and Mtb genetic associations with TB disease. The very fact that host variants in one population do not necessarily cross-validate across populations is evidence in support of population-specific adaptations. Specific Mtb lineages are likely to have co-evolved with distinct human populations. A key reference is missing (PMID: 23995134), which shows that different lineages co-evolved with human migrations. Also, meta-analyses of human GWAS studies to define variants associated with TB are very relevant to the topic of co-evolution (e.g., PMID: 38224499). eQTL studies can also highlight genetic variants associated with regulating key immune genes involved in the response to TB. The authors do mention that Mtb itself is relatively clonal with ~2K SNPs marking Mtb variation, much of which has likely evolved under the selection pressure of modern antibiotics. However, some of this limited universe of variants can still explain co-adaptations between distinct Mtb lineages and different human populations, as shown recently in the co-evolution of lineage 2 with a variant common in Peruvians (PMID: 39613754).

We thank the reviewer for these comments and agree we failed to cite and discuss the work from Sebastian Gagneux's group on co-migration, which we now discuss. We include a new paragraph discussing co-evolution as suggested on lines 145 – 155 and 218 -220 , citing the work proposed, which we agree enhances the arguments about co-evolution.

Although the examples of anti-TNF and anti-PD1 treatments are relevant as drivers of TB in limited clinical contexts, the bigger picture is that they highlight major distinct disease endotypes. These restricted examples show that TB can be driven by immune deficiency (as in the case of anti-TNF, HIV, and malnutrition) or hyperactivation (as in the case of anti-PD1 treatment), but there are still certainly many other routes leading to immune suppression or hyperactivation. Considering the idea of hyper-activation as a TB driver, the apparent higher rate of recurrence in the H56 trial referenced in the review is likely due to immune hyperactivation, especially in the context of residual bacteria in the lung. These different TB manifestations (immune suppression vs immune hyperactivation) mirror TB endotypes described by DiNardo et al (PMID: 35169026) from analysis of extensive transcriptomic data, which indicate that it's not merely different routes leading to the same final endpoint of clinical disease, but rather multiple different disease endpoints. A similar scenario is shown in the transcriptomic signatures underlying disease progression in BCG-vaccinated infants, where two distinct clusters mirrored the hyperactivation and immune suppression phenotypes (PMID: 27183822). A discussion of how to think about translating the extensive information from system biology into treatment stratification approaches, or adjunct host-directed therapies, would be helpful.

We agree with the points made and that the two publications above further enhance the paper. We have added discussion of the different disease endpoints on line 65 - 67, the evidence regarding immune hyperactivation versus suppression in the vaccination study on lines 162 - 164, and expanded on the translational implications on lines 349 - 352.

Reviewer #3 (Public review):

Summary:

This perspective article by Reichmann et al. highlights the importance of moving beyond the search for a single, unified immune mechanism to explain host-Mtb interactions. Drawing from studies in immune profiling, host and bacterial genetics, the authors emphasize inconsistencies in the literature and argue for broader, more integrative models. Overall, the article is thought-provoking and well-articulated, raising a concept that is worth further exploration in the TB field.

Strengths:

Timely and relevant in the context of the rapidly expanding multi-omics datasets that provide unprecedented insights into host-Mtb interactions.

Weaknesses (Minor):

Clarity on the notion of a "unified mechanism". It remains unclear whether prior studies explicitly proposed a single unifying immunological model. While inconsistencies in findings exist, they do not necessarily demonstrate that earlier work was uniformly "single-minded". Moreover, heterogeneity in TB has been recognized previously (PMIDs: 19855401, 28736436), which the authors could acknowledge.

We accept this point and have toned down the language, acknowledging that we are expanding on an argument that others have made, whilst focusing on the implications for the systems immunology era, and cite the previous work as suggested.

Evolutionary timeline and industrial-era framing. The evolutionary model is outdated. Ancient DNA studies place the Mtb's most recent common ancestor at ~6,000 years BP (PMIDs: 25141181; 25848958). The Industrial Revolution is cited as a driver of TB expansion, but this remains speculative without bacterial-genomics evidence and should be framed as a hypothesis. Additionally, the claim that Mtb genomes have been

conserved only since the Industrial Revolution (lines 165-167) is inaccurate; conservation extends back to the MRCA (PMID: 31448322).

Our understanding is that the evolutionary timeline is not fully resolved, with conflicting evidence proposing different dates. The ancient DNA studies giving a timeline of 6,000 years seem to oppose the evidence of evidence of Mtb infection of humans in the middle east 10,000 years ago, and other estimates suggesting 70,000 years. Therefore, we have cited the work above and added a sentence highlighting that different studies propose different timelines. We would propose the industrial revolution created the ideal societal conditions for the expansion of TB, and this would seem widely accepted in the field, but have added a proviso as suggested. We did not intent to claim that Mtb genomes have been conserved since the industrial revolution, the point we were making is that despite rapid expansion within human populations, it has still remained conserved. We therefore have revised our discussion of the conservation of the Mtb genomes on lines and 72 – 74, 81 – 83 and 185 – 190.

Trained immunity and TB infection. The treatment of trained immunity is incomplete. While BCG vaccination is known to induce trained immunity (ref 59), revaccination does not provide sustained protection (ref 8), and importantly, Mtb infection itself can also impart trained immunity (PMID: 33125891). Including these nuances would strengthen the discussion.

We have refined this section. We did cite PMID: 33125891 in the original submission but have changed the wording to emphasise the point on line ...

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Abstract

Line 30: What is an immunological route? Suggest

"...host-pathogen interaction, with diverse immunological processes leading to TB disease (10%) or stable lifelong association or elimination. We suggest these alternate relationships result from the prolonged co-evolution of the pathogen with humans and may even confer a survival advantage in the 90% of exposures that do not progress to disease."

Thank you, we have reworded the abstract along the lines suggested above, but not identically to allow for other reviewer comments.

Introduction

Ln 43: It is misleading to suggest that the study of TB was the leading influence in establishing the Koch's postulates framework. Many other infections were involved, and Jacob Henle, one of Koch's teachers, is credited with the first clear formulation (see Evans AS. 1976 THE YALE JOURNAL OF BIOLOGY AND MEDICIN PMID: 782050).

We have downplayed the language, stating that TB "contributed" to the formulation if Koch's postulated.

Ln 46: While the review rightly emphasises intracellular infection in macrophages, the importance and abundance of extracellular bacilli should not be ignored, particularly in transmission and in cavities.

We agree, and have added text on the importance of extracellular bacteria and transmission.

Ln: 56: This is misleading as primary disease prevention is implied, whereas the vaccine was given to individuals presumed to be already infected (TST or IGRA positive). Suggest

..."reduces by 50% progression to overt TB disease when given to those with immunological evidence of latent infection.

Thank you, edit made as suggested

Ln 62: Not sure why it is urgent. Suggest "high priority".

Wording changed as suggested.

Figure 1 needs clarification. The colour scale appears to signify the strength or vigour of the immune response so that disease is associated with high (orange/red) or low (green/blue) activity. The arrows seem to imply either a sequence or a route map when all we really have is an association with a plausible mechanistic link. They might also be taken to imply a hierarchy that is not appropriate. I'm not sure that the X-rays and arrows add anything, and the rectangle provides the key information on its own. Clarify please.

We have clarified the figure legend. We feel the X-rays give the clinical context, and so have kept them, and now state in the legend that this is highlighting that there are diverse pathways leading to active disease to try to emphasise the point the figure is illustrating.

Ln 149-157: I agree that the current dogma is that overt pulmonary disease is required to spread Mtb and fuel disease prevalence. It is vitally important to distinguish the spread of the organism from the occurrence of disease (which does not, of itself, spread). However, both epidemiological (e.g. Ryckman TS, et al. 2022Proc Natl Acad Sci U S A:10.1073/pnas.2211045119) and recent mechanistic (Dinkele R, et al. 2024iScience:10.1016/j.isci.2024.110731, Patterson B, et al. 2024Proc Natl Acad Sci U S A:10.1073/pnas.2314813121, Warner DF, et al. 2025Nat Rev Microbiol:10.1038/s41579-025-01201-x) studies indicate the importance of asymptomatic infections, and those associated with sputum positivity have recently been recognised by WHO. I think it will be important to acknowledge the importance of this aspect and consider how immune responses may or may not contribute. I regard the view that Mtb is an obligate pathogen, dependent on overt pTB for transmission, as needing to be reviewed.

We agree that we did not give sufficient emphasis to the emerging evidence on asymptomatic infections, and that this may play an important part in transmission in high incidence settings. We now include a discussion on this, and citation of the papers above, on lines 168 – 170.

Ln 159: The terms colonise and colonisation are used, without a clear definition, several times. My view is that both refer to the establishment and replication of an organism on or within a host without associated damage. Where there is associated damage, this is often mediated by immune responses. In this header, I think "establishment in humanity" would be appropriate.

We agree with this point and have changed the header as suggested, and clarified our meaning when we use the term colonisation, which the reviewer correctly interprets.

Ln 181-: I strongly support the view that Mtb has contributed to human selection, even to the suggestion that humanity is adapted to maintain a long-term relationship with Mtb

Thank you, and we have expanded on this evidence as suggested by other reviewers.

Ln 189: improved.

Apologies, typo corrected.

Figure 2: I was also confused by this. The x-axis does not make sense, as a single property should increase. Moreover, does incidence refer to incidence in individuals with that specific balance of resistance and susceptibility, or contribution to overall global incidence - I suspect the latter (also, prevalence would make more sense). At the same time, the legend implies that those with high resistance to colonisation will be infrequent in the population, suggesting that the Y axis should be labelled "frequency in human population". Finally, I can't see what single label could apply to the X axis. While the implication that the majority of global infections reflect a balance between the resistance and susceptibilities is indicated, a frequency distribution does not seem an appropriate representation.

The reviewer is correct that the X axis is aiming to represent two variables, which is not logical, and so we have completely changed this figure to a simple one that we hope makes the point clearly and have amended the legend appropriately. We are aiming to highlight the selective pressures of Mtb on the human population over millennia.

Ln 244: Immunological failure - I agree with the statement but again find the figure (3) unhelpful. Do we start or end in the middle? Is the disease the outside - if so, why are different locations implied? The notion of a maze has some value, but the bacteria should start and finish in the same place by different routes.

We are attempting to illustrate the concept that escape from host immunological control can occur through different mechanisms. As this comment was just from one reviewer, we have left the figure unchanged but have expanded the legend to try to make the point that this is just a conceptual illustration of multiple routes to disease.

Ln 262 onward: I broadly agree with the points made about omic technologies, but would wish to see major emphasis on clear phenotyping of cases. There is something of a contradiction in the review between the emphasis on the multiplicity of immunological processes leading ultimately to disease and the recommendation to analyse via omics, which, in their most widely applied format, bundle these complexities into analyses of the humoral and cellular samples available in blood. Admittedly, the authors point out opportunities for 3-dimensional and single-cell analyses, but it is difficult to see where these end without extrapolation ad infinitum.

We totally agree that clear phenotyping of infection is critical, and expand on this further on lines 307 - 309.

Reviewer #2 (Recommendations for the authors):

I suggest expanding on the genetic determinants of Mtb/host co-evolution.

Thank you, we have now expanded on these sections as suggested.

Reviewer #3 (Recommendations for the authors):

We are in an era of exploding large-scale datasets from multi-omics profiling of Mtb and host interactions, offering an unprecedented lens to understand the complexity of the host immune response to Mtb-a pathogen that has infected human populations for thousands of years. The guiding philosophy for how to interpret this tremendous volume of data and what models can be built from it will be critical. In this context, the perspective article by Reichmann et al. raises an interesting concept: to "avoid unified immune mechanisms" when attempting to understand the immunology underpinning host-Mtb interactions. To support their arguments, the authors review studies and provide evidence from immune profiling, host and bacterial genetics, and showcase

several inconsistencies. Overall, this perspective article is well articulated, and the concept is worthwhile for further exploration. A few comments for consideration:

Clarity on the notion of a "unified mechanism". Was there ever a single, clearly proposed unified immunological mechanism? For example, in lines 64-65, the authors criticize that almost all investigations into immune responses to Mtb are based on the premise that a unifying disease mechanism exists. However, after reading the article, it was not clear to me how previous studies attempted to unify the model or what that unifying mechanism was. While inconsistencies in findings certainly exist, they do not necessarily indicate that prior work was guided by a unified framework. I agree that interpreting and exploring data from a broader perspective is valuable, but I am not fully convinced that previous studies were uniformly "single-minded". In fact, the concept of heterogeneity in TB has been previously discussed (e.g., PMIDs: 19855401, 28736436).

We accept this point, and that we have overstated the argument and not acknowledged previous work sufficiently. We now downplay the language and cite the work as proposed.

However, we would propose that essentially all published studies imply that single mechanisms underly development of disease. The authors are not aware of any manuscript that concludes "Therefore, xxxx pathway is one of several that can lead to TB disease", instead they state "Therefore, xxxx pathway leads to TB disease". The implication of this language is that the mechanism described occurs in all patients, whilst in fact it likely only is involved in a subset. We have toned down the language and expand on this concept on line 268 – 270.

Evolutionary timeline and industrial-era framing. The evolutionary model needs updating. The manuscript cites a "70,000-year" origin for Mtb, but ancient-DNA studies place the most recent common ancestor at ~6,000 years BP (PMIDs: 25141181; 25848958). The Industrial Revolution is invoked multiple times as a driver of TB expansion, yet the magnitude of its contribution remains debated and, to my knowledge, lacks direct bacterial-genomics evidence for causal attribution; this should be framed as a hypothesis rather than a conclusion. In addition, the statement in lines 165-167 is inaccurate: at the genome level, Mtb has remained highly conserved since its most recent common ancestor-not specifically since the Industrial Revolution (PMID: 31448322).

We accept these points and have made the suggested amendments, as outlined in the public responses. Our understanding is that the evidence about the most common ancestor is controversial; if the divergence of human populations occurred concurrently with Mtb, then this must have been significantly earlier than 6,000 years ago, and so there are conflicting arguments in this domain.

Trained immunity and TB infection. The discussion of trained immunity could be expanded. Reference 59 suggests the induction of innate immune training, but reference 8 reports that revaccination does not confer protection against sustained TB infection, indicating that at least "re"-vaccination may not enhance protection. Furthermore, while BCG is often highlighted as a prototypical inducer of trained immunity, real-world infection occurs through Mtb itself. Importantly, a later study demonstrated that Mtb infection can also impart trained immunity (PMID: 33125891). Integrating these findings would provide a more nuanced view of how both vaccination and infection shape innate immune training in the TB context.

We thank the reviewer for these suggestions and have edited the relevant section to include these studies.

<https://doi.org/10.7554/eLife.108175.2.sa4>